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METHOD

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Battlefield Triage Technology: Addressing Effectiveness in Far-Forward Combat Casualty Care

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OCCUPATIONAL APPLICATIONS

We identified strategies to support the development or evaluation of technology solutions for a medical response in remote and austere environments. We conducted a systematic analysis of relevant tasks and potential errors within battlefield medical response and communicated the resultant system taxonomy for interdisciplinary application by both engineers and health care practitioners. Our paper contributes to continued advancements of technology support for remote casualty care through detailed insights into remote medical response, identification of system vulnerabilities, and a consolidated system analysis.

TECHNICAL ABSTRACT

Background: During mass casualty incidents there is a scarcity of resources relative to the number of casualties. Medics acting in far-forward military settings must manage a mass casualty response and implement casualty care in dynamic, remote environments. Further integration of medical technology may support a medical response and improve casualty outcomes.

Purpose: Employ structured systems analyses of the complexity of the tasks, actors, and work domain to inform constraints of a technology intervention in far-forward casualty care.

Methods: Seven subject matter experts, drawn from diverse backgrounds and experiences within the Canadian Armed Forces medical trades, participated in the structured workshop. Hierarchical Task Analysis and Human Failure Analysis were applied during a 2-day workshop. Participants analyzed two scenarios considering different organizational units, environments, locations, and types of injuries. Perspectives on select current tools and systems were captured through the system usability score survey. Work Domain Analysis and Worker Competency Analysis were informed by further subject matter expert engagement and protocol documentation.

Results: Mass casualty medical response tasks were decomposed and then analyzed to identify sources and risks of human failure. Informational elements of the work domain associated with actions that introduce the most risk from potential human failure focused the worker competency analysis. The resultant system analysis yielded a matrix of skill, rule, and knowledge-based behaviors required of a system designed to support a medic during combat casualty care. This matrix is designed to guide or evaluate the design of a system to support medical field care.

Conclusions: Vulnerabilities attributed to information collection, derivation, and communication during field care can be mitigated through technology integration.

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KEYWORDS

Decision support; Remote medical response; Cognitive Work Analysis method

1. Introduction

A mass casualty event is defined as a scarcity of medical and logistic capacity for the number of casualties (Cicekci et al., 2022). Mass casualty (MASCAL) events necessitate sorting and prioritization for both medical intervention and evacuation from the point of injury. On the battlefield, casualties are first evacuated to a

‘casualty collection point’ (CCP). The CCP is an area secure from enemy assault and exists only while a secure perimeter is sustained. Medical interventions are first possible under ‘tactical field care’ in the CCP, but the planned medical response immediately reverts to ‘care under fire’ if the area becomes vulnerable to enemy combatants (Bhaumik et al., 2022; Falzone et al., 2017).

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During tactical field care, the casualty is first assessed for the most life-threatening injuries following the MARCHE mnemonic (Massive hemorrhage control, Airway management, Respiratory management, Circulation, Hypothermia, Head injury, Eye injury, and Everything else). The highest priority (MAR) evaluations are repeated until all casualties are stable. Importantly, even a life-threatening injury is not treated until the completion of the MAR survey, as it cannot be correctly prioritized over unknown injuries. Undiscovered life-threatening injuries severely worsen patient outcomes, particularly as equipment to reverse damage caused by prolonged inattention (e.g., blood transfusion) are not always available in the field. The medic maintains situational awareness of physiological signs relevant to the mechanism of injury for each casualty in the CCP through repeated casualty assessment to optimize treatment and evacuation prioritization.

Treatment priority is dependent on medical resources, health status, and expected injury course. Evacuation priority additionally considers available resources in upstream health facilities; evacuation platform viability, security, and capacity; and resource coordination across an operational theater (Cicekci et al., 2022). From the CCP, casualties may be further evacuated to facilities with more support and resources in or out of operational theater.

If evacuation is delayed beyond the anticipated (and trained for) window of time, the medical response enters 'prolonged field care'. This form of casualty management requires a departure from the rule-governed actions of prehospital medical procedures for an increased complexity of medical decisions made from casualty assessments and reassessments (Keenan & Riesberg, 2017; Pilgrim et al., 2023). Medics manage resource allocation and medical interventions with judgments based on understanding injury manifestation and predicted injury course (Cicekci et al., 2022). While complex physiological factors are integrated into decision making and management in hospitals, current in-field casualty assessments are restricted to signs and symptoms that can be observed or measured in lower resource settings with lower light and sound to avoid detection (Centre, 2023; Cicekci et al., 2022; Ryan et al., 1990). Medics must extrapolate information from vital signs to infer injury course and an appropriate treatment regimen. Many tasks associated with prolonged field care require much more physical dexterity, such as needle decompression, and cognitive load, such as analytical assessments of vital sign trends (e.g., urine output, mental status, loss of peripheral blood flow, cap refill,

lactate levels). Vital sign trends improve understanding of organ function and inform treatment decisions. Tools used for tracking this information are often a sheet of paper and marker (Centre, 2023; Cicekci et al., 2022). These tools are affected by adverse weather, information may be recorded with poor legibility or incorrectly due to haste or smearing, and text-based data may impede recognition of trends.

The limited ability to monitor physiological function, combined with the reduced options for interventions and increased risk of medical complications (e.g., shock or sepsis), worsen casualty health outcomes with longer time spent in low resource settings (Nemeth et al., 2023). Within the scope of this manuscript, our investigation focuses on technology that may be strategically integrated to support casualty management in far forward positions. The accumulated progress from technical fields of silicon manufacturing, antenna design, communication protocols, and display elements across the past decades has improved the potential efficacy to integrate analytical and informational support for far forward medical responses (Ahmad et al., 2023; Aledhari et al., 2022; Bazyar et al., 2020; Cassagnol et al., 2019; Olivia et al., 2021).

There exists the technological capacity to measure, analyze, store, and communicate physiological signals to inform the health status of a casualty (Tharion & Kaushik, 2006). Examples of such signal measure, communication, and display technology has been effectively integrated in provisional medical settings to support remote patient monitoring (Steinhubl et al., 2016). However, despite a long established popularity and diversity of approaches represented in literature and patents, there remains a lack of increased technology use during combat casualty care (Nemeth et al., 2023; Sarma et al., 2023; Schmidt et al., 2023). Technology integration into complex systems requires consideration beyond technology capacity. Any intervention should include consideration for the extreme environment and cognitive states in which tactical field care and prolonged field care are performed.

Hierarchical Task Analysis (HTA) and Human Failure Analysis (HFA) capture goals and details of steps by which tasks are achieved. Researchers have previously employed HTA and HFA in healthcare to investigate challenges in procedures and protocols, such as continuity of care during patient transfer or planning for an emergency response (Annett & Duncan, 1967; Benton et al., 2023; Bourne et al., 2022; Jonson et al., 2017; Raduma-Tomás et al., 2012).

Cognitive Work Analysis (CWA) was introduced to model the interactions between humans and

information within complex socio-technical systems through formative analysis (Vincente, 1999). Work Domain Analysis (WDA) and Worker Competency Analysis (WCA) are the first and last phases of CWA, respectively, that yield the Ecological Interface Design (EID) framework when applied sequentially. EID identifies characteristic information elements which best support cognitive demands imposed on a worker acting within a system to achieve its goals (Austin et al., 2022; Burns, 2000; Burns & Hajdukiewicz, 2004; Naikar, 2013; Naikar & Ben, 2021). WDA builds a hierarchical relationship linking system goals to the function of its physical components. The WDA analysis decomposes a system into elements across levels of information abstraction and physical decomposition to analyze functional constraints imposed by the work domain (Naikar, 2013). Communicating knowledge of a system's structure supports a worker to generalize and more efficiently process information from the system (Rasmussen, 1985).

WCA considers the behavior of actors operating within the established work domain constraints. The different types of behaviors—specifically skill (SBB), rule (RBB), and knowledge-based (KBB)—describe the different levels of information and processing required to produce the behavior. Each type of behavior may be necessary (Naikar & Ben, 2021). Most operational contexts require medics to respond to information through reaction (i.e., SBB) or a defined protocol (i.e., RBB). However, behaviors during a novel event, such as those that arise during the complexity of MASCAL events, cannot be planned, thus require an understanding of the CCP system to determine a course of action (i.e., KBB). The outcomes from this analysis identifies how the three types of behaviors can be supported through interface design to optimize human cognition (Kilgore & St-Cyr, 2006). We applied EID to explore the information required to support actions of medics within the constraints of a CCP system.

In the following, we present our analysis of mass casualty incidents using human factors frameworks to identify capabilities and constraints required of technology that would be integrated in a remote medical response. Our initial analytical representation of the CCP through HTA examines the system constraints and tasks that support system goals. Risk and categorization of potential failures from the HFA informs information elements with the greatest potential to be supported by increased technical intervention. The culmination of our analyses with the WCA translates system vulnerabilities to required behavioral competencies and, consequently, technical functions. Our objective for this research was to outline key

considerations for a system designed to support medics during far-forward combat casualty care.

2. Methods

Data were collected during a two-day data collection activity termed 'workshop' (described below in section 2.1). We invited seven volunteers of a combination of officers and noncommissioned members, with ranks of Master Corporal to Major, including five males, with mean (SD) ages of 35 (6) years, who were recruited from four different units of the Canadian Armed Forces with diverse and varied experiences in the medical trades. This research was approved by the Defence R&D Canada Human Research Ethics Committee, and all participants provided consent. Understanding of the CCP was built through exploratory research opportunities additional to the aforementioned data collection activity; we observed live practical medic trainee evaluations and participated in high-fidelity mass casualty simulations. Additionally, we referenced medical technician protocols and procedures established by the Canadian Forces Health Services (Cicekci et al., 2022; Kilgore & St-Cyr, 2006; Naikar, 2013). The data collection activity, reference material, and exploratory opportunities informed our application of WDA and WCA to the system domain of the CCP to identify information to best support cognitive demands of a medic performing tactical field care in a mass casualty scenario.

2.1. Workshop Design

The workshop framed discussions around two MASCAL scenarios (listed below) which provided context for evaluating procedures and resources that would be employed.

1. An artillery strike produced four casualties: 1. Penetrating shrapnel injury to the lower abdomen and thighs. 2. Arm amputation 3 & 4. Blunt force trauma to chest and head. The donned kit was a fragmentation vest and helmet. Weather is not a factor, but the environment is exposed and remote. A single medic is the medical authority.
2. A motor vehicle accident produced five casualties: 1, 2 & 3 partial thickness burns to face and neck, and a full thickness burn to forearms 4 & 5 blunt injury to right side of the body with penetrating shrapnel to arms and legs. Casualties are being received at a unit medical station on a forward operating base in winter

weather. The responding medical personnel are a medical officer, physician assistant, and two medics.

Our research team guided participants to identify every task and action they would perform in each scenario through a “walk-through-talk-through” performed in two groups (divided by rank to support open discussion). We followed the procedure outlined in Benton et al. (2023). We then analyzed the participants’ reports through Hierarchical Task Analysis (HTA) and Human Failure Analysis (HFA) frameworks. During the workshop, multiple small and full group discussions were held to engage participants throughout the HTA and HFA analyses (Annett & Duncan, 1967; Benton et al., 2023; Bourne et al., 2022; Jonson et al., 2017). We employed the System Usability Scale (SUS; Tullis & Albert, 2013) during the final day of the workshop to evaluate the usability of tools and technologies currently used during a MASCAL medical response. The tools and technologies evaluated were identified by participants during the HTA as being employed during the response to the two scenarios. The survey comprised 10 questions which queried effectiveness, efficiency, and satisfaction (Tullis & Albert, 2013).

3. Results

The HTA yielded a comprehensive list of operational actions structured by overarching tasks and their dependencies. Participants identified select tasks as especially relevant considering vulnerability of the task to failure, sensitivity of system to the task failure, and areas tolerant to system intervention. Prioritized tasks included the following: performing a triage sweep on a casualty using the START (Simple Triage And Rapid Treatment) algorithm, identifying triage priority, and labeling casualties with the corresponding colored tag (i.e., from highest to lowest priority of treatment: immediate—red tag, delayed—yellow tag, minimal—green tag, and expectant—black tag), establishing key locations within the casualty collection point, planning and maintaining communication flow to coordinate evacuation logistics, identifying evacuation priority of casualties from the casualty collection point, performing rapid physical examination to identify less visible immediate injuries, and performing a more thorough physical exam to diagnose and track physiologic trends that may manifest during prolonged patient care. A summary of the human failure analysis (HFA) performed for each of the prioritized tasks is shown in Table 1.

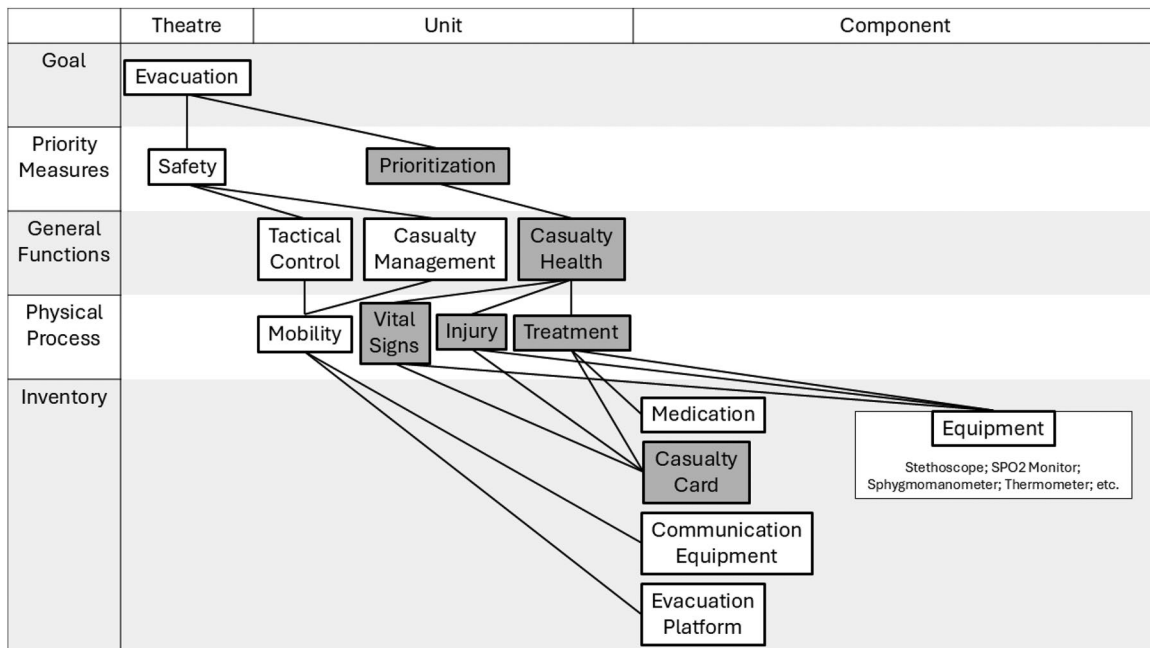
Within the CCP, actors operate at the level of ‘theater of operation’, with strategies employed across all combat in the battlefield. ‘Theater of operation’ can be broken down broadly into organized operation of units, and, further, an inventory of physical components. The physical components in play at the CCP include communication equipment handled by the signaler or second in command, vehicle platforms (air or land) used to evacuate casualties, medications/medical resources consumed through casualty treatment, the casualty card (a hand-sized document used to record information), and equipment carried by the medic that is not consumed with treatment. Physical system components support movement of casualties (i.e., mobility), measure and track vital signs, identify and track injuries, and document the performed treatment. Information of the system elements are abstracted to levels of physical process, general function, priority measure, and goal. The general functions of the physical processes support safety and prioritization measures through maintenance of tactical control, casualty management, and knowledge of relative health status of casualties. Figure 1 summarized results of the WDA in the abstraction-decomposition (AD) model of a CCP work domain. As previously explained, these prioritized tasks (identified through workshop discussions) are those performed by the medic whose failure represents the highest risk to the medical response within the CCP.

We chose to focus the WCA on the competencies required for acting within the domains most vulnerable to failure (see Table 1 and shaded calls in Figure 1). Our WCA structured a taxonomy of behaviors required to support a medic’s actions within the CCP into a skill-, rule-, knowledge behavioral inventory (SRK inventory). Results of our WCA of a CCP are shown in Table 2.

Table 3 shows the results of the application of the SUS to current equipment and protocols verbally “employed” by the participants during the workshop scenarios. Both protocols and equipment were analyzed. The MARCHE mnemonic (previously described) directs the process for assessing casualties that occurs periodically during tactical and prolonged field care. The MIST-AT mnemonic (Mechanism, Injury, Signs, Treatment -Age, Time) identifies the used to pass on casualty information during a casualty ‘hand-off’ during evacuation. The SMART Triage Tag™ (Kingfisher Medical, 2020), a vital trend chart, and the Tactical Combat Casualty Care Casualty Card (TCCC Card; Kotwal et al., 2013) are different forms of a printed template carried by every service member that medics use to record information of the injured service member and any treatment provided. The blood pressure

Table 1. Human failure analysis of tasks selected from the hierarchical task analysis.

Task	Failure risk	Failure mode	Influence factor	Control measures
Apply START triage algorithm	Over- or under-triage	Rule based mistake	Number of casualties High stress	N/A
Direct casualties to CCP	Med tech overrode in CCP management	Routine violation	Number of casualties	N/A
Establish evacuation plan	Lack of situational awareness of evacuation platform leading to a mismanagement of consumable medical resources	Situational violation	Lack of practice in combat environment	N/A
Prioritize evacuation	Error in calculating respiration or pulse rate Health status not recognized Health status not recorded effectively	Rule-based mistake Knowledge- based mistake Rule-based mistake OR routine violation	Equipment failure Medic view obscured Injury presentation not recognized Adherence to START algorithm	Patient vital sign assessment repeated every 5 min if possible.
Prepare casualty for evacuation	Casualty leaves CCP without knowledge of medic Casualty leaves CCP in incorrect order	Rule-based mistake Rule-based mistake	Person out of loop during personnel shifts	Established choke point in CCP
Perform physical exam to identify fatal injuries	Life-threatening injury unidentified	Rule-based mistake	Casualty cover in extreme cold Cognitive function	Casualty card
Perform detailed physical exam and act to support casualty	Injury presentations overlooked Treatment applied incorrectly Information not recorded effectively Relevant trends and patterns may be missed Chart matched to wrong casualty	Knowledge-based mistake Knowledge-based mistake Slip of action Lapse of memory Systemic routine violation Knowledge-based mistake Slip of action	Long evacuation window Cognitive function Precision of interventions Space in MIST card Weather Stress Fatigue Multiple care providers	Reference materials Casualty card

**Figure 1.** Abstraction Decomposition model of the CCP work domain. Shaded elements highlight the elements of the CCP system within which key failures occur for medics and represent domains in which the prioritized tasks of the HTA occur (see Table 1).

cuff, stethoscope, and SpO₂ finger clip are diagnostic tools common in civilian medicine used to measure blood pressure, listen to heart sounds, and measure

blood oxygenation respectively. The Propaq M is a compact vital signs monitor (ZOLL Medical Corporation, 2025).

Table 2. Skill, rule, knowledge behavioral inventory for possible medic actions associated with each information element highlighted in the AD.

Information element	Measure	Resultant state	Skill-based behavior	Rule-based behavior	Knowledge-based behavior
Casualty prioritization	Direct perception of the color of casualty card	Relative order for casualty evacuation	Recognize color of casualty card	Know tiered prioritization classes	Understand relative severity of casualty health status
Casualty health status	Derived from casualty information	Need for medical intervention	Recognize vital sign trends	Associate correct protocols with vital signs, injury information, and previous treatment	Understand physiological etiology associated with measured signals and intervention response
Vital signs	Derived by the medic or measured with sensor (SPO2)	State of physiological function	Recognize value of vital sign	Know the value of measure relative to threshold	Understand state of physiology indicated by vital signs
Injury	Directly perceived	Mechanism, severity, location, and presentation of injury	Locate any injuries	Associate protocols with injury type	Insight into underlying systems impacted by injury
Treatment	Directly perceived	Treatment applied to casualty	Identify treatment noted on casualty card	Associate medical resources and procedures with treatment	Knowledge of pharmacological effect of medication

Highlighted cells indicate information that is currently communication through existing tools (i.e. the casualty card).

Table 3. Mean (and standard deviation) across participants in SUS score evaluation of medical equipment and protocols applied during MASCAL and triage.

	Item	SUS mean (std)
Protocol	MIST-AT	83 (7.5)
	MARCHE	70 (15.2)
Equipment	SMART Triage Tag™	77 (10.0)
	Vital trend chart	76 (10.5)
	Blood pressure cuff	80 (18.4)
	SpO ₂ finger clip	95 (9.6)
	TCCC card	84 (17.1)
	Propaq M	66 (18.7)
	Stethoscope	89 (10.4)

4. Discussion

Our Human Failure Analysis (HFA) highlighted modes of failure experienced during a MASCAL medical response, factors that impact failures, and the estimated risks of each failure. Our identified failures align with capabilities essential to prolonged field care identified by Keenan and Riesberg (2017). Technology intervention is best applied to inadvertent errors attributed to action, judgment, or decision making. Changes to training and policies, conversely, address deliberate system failures such as routine, situational, or exceptional violations (Benton et al., 2023).

4.1. Prioritized Care Under Fire Tasks

A ‘triage sweep’ is an individual task performed applying the START algorithm to assign the casualty to a triage category in less than 30s. The most common failure during care-under-fire is over triage (i.e., assigned higher priority than necessary; Lupton et al., 2023). This failure unnecessarily burdens capacity at all subsequent levels of care. Under-triage can also

occur, especially if the casualty is assessed as “Expectant”. In this case, casualties are prioritized lower than is appropriate and will not receive needed medical treatment. Lupton et al. (2023) found that the best procedures when evaluated by under-triage rate had intolerably high over-triage rates (>60%). Ideally, the triage process yields less than 30% over-triage (casualties mistakenly assigned too high a prioritization) and less than 5% under-triaged (casualties assigned too low a prioritization) (Bhaumik et al., 2022). Experience level most prominently affects likelihood of mis-triage. Increased volume of casualties increases the likelihood of over-triage; medical technicians err on the side of caution when their assessment time is more limited. Due to the rapid nature of this phase of the medical response, it is unlikely technology can be integrated successfully to support the medic’s actions and decision making.

4.2. Current Armamentarium within Tactical Field Care

Discussions held to formulate the HTA and HFA during the workshop provide the context for which the SUS results are discussed. The SpO₂ figure clip is the most highly rated item; it alone provides a single digital value output. This form of information presentation differs from the heart rate and blood pressure, which the medic must derive using medical equipment (i.e., through auscultation and sphygmomanometry respectively). Interestingly, the Propaq portable vital signal monitor provides continuous monitoring of biological signals, yet participants rate the Propaq at the lowest system usability during far

forward combat casualty care with a SUS score of 66 (Table 3; Tullis & Albert, 2013). These results suggest technological integration during tactical field care at the CCP is prioritized by effective information communication, ease of use, and associated load carriage. During a far-forward combat casualty care response, a detachment-level medic initially relies on medication and equipment from within a body-borne pack. Carriage weight and volume are scarce resources and so it follows that heavy or bulky items (such as the Propaq) are disfavored.

Medical providers use a casualty card to note injuries, treatments, and vital sign values. The two casualty cards evaluated through SUS were the SMART Triage Tag™ and the TCCC Card (see Table 3). While the color of the SMART Triage Tag™ is helpful to quickly identify triage category, participants noted how it does not prioritize information used for tactical field care or to be communicated during casualty hand-off. The SMART Triage Tag™ was also noted to not have sufficient space for prolonged recording of vital sign measures. Further, discrete values (e.g., with a BP cuff, stethoscope, or SpO₂ monitor) require extrapolation to recognize relevant trend information (Ahmad et al., 2023; Nemeth et al., 2023). Nelson et al. (2022) similarly recognized design elements of paper casualty cards that pose barriers encountered during casualty management. The TCCC card is implemented in place of the SMART Triage Tag™ during some operations.

There exists a fundamental difference in the systems for which the two tools were designed. While the SMART Triage Tag™ was designed for civilian implementation, the TCCC card was designed specifically for tactical combat casualty care and emphasizes information especially relevant in a military context (Bhaumik et al., 2022; Cicekci et al., 2022; Cone & MacMillan, 2005). The TCCC card mirrors the flow of information requested in a casualty hand-off during evacuation, includes prompts for information relevant to military theater (e.g., identification, type of injuries, and type of treatments), and structures space for recording vital sign values. Streamlining information to support evacuation tasks is valuable to ensure safe continuation of care. Despite the difference between the participants' usability ranking of the SMART Triage Tag™ and TCCC card (SUS scores of 76 and 84, respectively; see Table 3) there were no significant (signed rank statistic = 5, $\alpha > 0.2$) differences in participants' perspective on the tools' usability. However, the infrequent use of the TCCC card contributed to an

increased variation in participant perception of use (see standard deviations of Table 3).

More ubiquitous experience with the TCCC card would better align participant experience and provide a more accurate comparison of usability with the SMART Triage Tag™. Participants of the workshop were selected to prioritize different experiences and with medical response. Thus, our recruitment goal of participants with diverse experiences to support generalization of the HTA and HFA resulted in high levels of standard deviation in SUS results.

4.3. Human Factors Indication for Technology Intervention

There are few existing safeguards established during tactical field care (see Table 1). Consequently, failure mitigation predominantly relies on re-assessment or repetition of tasks. Indeed, in monitoring casualties at the CCP, the only control measures for casualty prioritization are the repetition of medical assessments which incurs costs in lost treatment time, potentially lost evacuation opportunity, and a high burden to periodically reassess casualties. Technology integration to support medic actions within the CCP should redirect the burden of measurement and information integration from the medical personnel and communicate information necessary to support worker competencies. The WCA (Table 2) identified the skilled reactions, learned procedures, and critical thinking with which medics respond to information during tactical field care.

Continuous biological signal monitoring enabled through a sensor system reduces failure vulnerability from mental rate derivation, eliminates the need for reapplication of an assessment, and provides trending information more sensitive to changes in health status (Benjamin et al., 2023; Borsotto et al., 2004; Cancio et al., 2008; Steinhubl et al., 2016). These measures reduce cognitive load imposed on the medic and can be communicated through an interface to support skill-based reactions. Further, medical resources and established protocols can be easily made available to support rule-based actions with the introduction of a digital user interface. Incorporating analytics and machine learning to biological signal reception can recommend medical procedures associated with vital sign information. Cumulation of physiological signal information may be analyzed to infer relative health statuses of casualties, identify risk of organ decompensation, or suggest hidden injury patterns. Examples in prior research have shown that the presence of an unseen life-threatening injury can be identified

through multifactor integration of biological signal information and communicates information of underlying systems impacted by injury (Benjamin et al., 2023; Jin et al., 2024). Analytics and modeling can provide insight into physiological etiology and support knowledge-based decision making.

The WCA culminates our analysis of vulnerabilities, processes, and functional constraints of tactical field care within a CCP into a SRK inventory. We encourage the application of the analysis as a first step to evaluate potential technology interventions or to guide system design decisions. A case study demonstrating an example application of our analysis to evaluate a medical decision support design is shown below.

The Battlefield Assisted Trauma Distributed Observation Kit (BATDOK) is an android application designed and implemented by the American military to facilitate wireless monitoring of multiple casualties' vitals simultaneously and continuously (Sarma et al., 2023). BATDOK allows for a casualty's medical information to be updated by the medic directly or through scanning optical data representations (e.g., QR code; Burnett, 2018). The system depends on integration with signals from sensor networks placed on the casualties. The device also supports information transfer between devices (Burnett, 2018). While clear document transfer is essential for patient handoff and continuity of care during evacuation, during tactical field care it can also be used to maintain situational awareness between medical providers within the CCP.

BATDOK better communicates casualty information to the medic during tactical field care. The column 'System Measure' compares the measurement of each

information element with the current practice and indicates if the system offers and improvement (+) or not (-). Outlined cells indicate behaviors supported by the interface (see Table 4). Relaying treatment records and casualty prioritization information through the interface supports the medic's ability to compare and verify casualty information remotely. Supporting sensor measurement of vital signs reduces the burden of casualty reassessment and improves information relevancy. However, medic actions to identify casualty injuries and determine current casualty health statuses are unchanged. All skill- and rule-based behaviors associated with key informational elements are supported by the interface. Reflected by the low support for knowledge-based behavior, a limitation of the BATDOK is the lack of information integration and analytics.

4.4. Limitations and Future Research

Despite reviewing two distinct environments during the workshop, our casualty scenario analyses were restricted by participant experience and information available to support scenario description. While participants were experts in the field of combat casualty care, the inclusion of only seven subject matter experts limits experiences and restricts the external validity of our findings. Thus, it is important to note that the HTA, HFA, and SUS reflect the context of the two scenarios as the environment of the casualty dictates many aspects of potential mechanisms of injuries, medical intervention, and response in a MASCAL event. While most tools and protocols remain, casualty

Table 4. Application of the skill, rule, knowledge behavior inventory to evaluate the BATDOK system.

Information element	Current measure	System measure	Skill-based behavior	Rule-based behavior	Knowledge-based behavior
Casualty prioritization	Direct perception of the color of casualty card	Direct perception of casualty color on interface (+)	Recognize color of casualty card	Know tiered prioritization classes	Understand relative severity of casualties' health status
Casualty health status	Derived from casualty information	Derived from casualty information (-)	Recognize vital sign trends	Associate correct protocols with vital signs, injury information, and previous treatment	Anticipate physiological etiology associated with measured signals and intervention response
Vital signs	Derived by the medic or measured with sensor (SPO2)	Directly perceived on interface and measured by sensor (+)	Recognize value of vital sign	Know the value of measure relative to threshold	Understand state of physiology indicated by vital signs
Injury	Directly perceived	Directly perceived (-)	Locate any injuries	Associate protocols with injury type	Insight into underlying systems impacted by injury
Treatment	Directly perceived	Directly perceived on interface (+)	Identify treatment noted on casualty card	Associate medical resources and procedures with treatment	Knowledge of pharmacological effect of medication
Total		+3	100%	100%	20%

The column 'System Measure' compares the measurement of each information element with the current practice and indicates if the system offers and improvement (+) or not (-). Highlighted cells indicate behaviors supported by the interface.

transportation on a ship, for example, is logistically different than that on ground (Butler et al., 2022). Similarly, operation within the Arctic environments necessitates accommodating extreme cold temperatures and remote isolation during tactical field care (Samblanet & Booy, 2022). Further, our analyses focused on intervention supported through technology. However, failures for which mitigation could be achieved through a change in training or operational policy were identified. Future research should endeavor to investigate a greater number of subject matter experts and experiences not currently represented in our participant pool (i.e., experiences in air evacuation, ships, and the arctic).

5. Conclusion

Employing technology to reduce the burden of information collection, derivation, and communication will support the medical response in mass casualty triage. However, the complexity of the tasks, environment, and actors within the CCP necessitates a systematic consideration of technology intervention. We condense the results of our analysis of casualty management into the SRK inventory to present recommendations of the incorporation of human factors into the design of a technology system to support far-forward tactical field care.

Conflict of Interest

The authors declare no conflict of interest.

Author contributions

CRedit: **Kaela Shea**: Conceptualization, Data curation, Formal analysis, Methodology, Supervision, Writing – original draft, Writing – review & editing; **Elizabeth Wong**: Data curation, Formal analysis, Methodology, Writing – review & editing; **Craig Burrell**: Data curation, Investigation, Methodology, Supervision, Writing – review & editing.

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